

ABOUT ME

Data Scientist and AI Engineer with hands-on experience delivering data-driven and AI-based solutions across industry, healthcare, research, and technology domains. Experienced in working across the full lifecycle of data and AI systems, including large-scale data analysis, visualization, pipeline development, and the design and deployment of production-grade AI solutions such as intelligent automation, document processing systems, and domain-specific chatbots. Proven ability to collaborate with cross-functional teams, translate complex problems into practical solutions, and operate effectively in both structured corporate environments and fast-paced, applied settings. Analytical, adaptable, and detail-oriented, with a strong focus on reliability, scalability, and measurable impact.

EDUCATION AND TRAINING

15/05/2021 – 15/05/2025 Swabi, Pakistan
BACHELORS OF DATA SCIENCE Ghulam Ishaq Khan Institute

Website <https://giki.edu.pk> | Level in EQF EQF level 8

01/01/2019 – 01/04/2021 Islamabad, Pakistan
A-LEVELS Roots IVY F-8 CampusEngi

Website <https://ivycollege.com.pk> | Field of study Engineering and engineering trades not further defined | Level in EQF EQF level 8

WORK EXPERIENCE

DATA SCIENTIST & AI ENGINEER – OIL & GAS DEVELOPMENT COMPANY LIMITED (OGDCL) – 01/01/2026 – Current – ISLAMABAD , PAKISTAN

I am currently responsible for cleaning, processing, and validating large-scale operational and production datasets to ensure high data quality and reliability for downstream analysis. My role involves performing exploratory data analysis and feature engineering to extract actionable insights, as well as designing and developing clear, impactful visualisations and dashboards to support data-driven decision-making for both technical and management stakeholders. I collaborate on data engineering initiatives, including the development and maintenance of data pipelines, transformations, and integrations across multiple internal data sources. In addition, I work as an AI Engineer within a dedicated AI team, contributing to the development of a personalised organisational chatbot by supporting its design, data preparation, and model integration to deliver accurate, contextual, and domain-specific responses. I work closely with cross-functional teams, including engineers and domain experts, to ensure that analytical and AI solutions are aligned with overall business objectives.

AI ENGINEER – BELLMEDEX – 28/08/2025 – Current – RAWALPINDI, PAKISTAN

Lead AI initiatives to design and implement intelligent systems that automate and optimize complex workflows. Experienced in developing and deploying solutions across computer vision, machine learning, AI-driven automation, and workflow orchestration using n8n, delivering scalable, production-ready systems that improve efficiency, accuracy, and operational performance. Collaborate with cross-functional teams to translate business and domain challenges into practical AI solutions, ensuring reliability, maintainability, and measurable impact.

AI ENGINEER INTERN – NECOP – 25/04/2025 – 25/08/2025 – ISLAMABAD, PAKISTAN

Worked on **applied AI research and development**, focusing on **computer vision and deep learning-based automation systems**. Assisted in designing, training, and evaluating **custom neural networks** for object detection and classification tasks. Collaborated with senior engineers to **optimize model performance** and **deploy inference pipelines** in real-world environments. Gained practical experience in **data preprocessing, model tuning, and experimentation**, contributing to projects aimed at improving **accuracy, efficiency, and scalability** of AI-driven solutions.

VICE-PRESIDENT – GHULAM ISHAQ KHAN INSTITUTE SPORTS SOCIETY – 10/01/2022 – 10/07/2025 – SWABI, PAKISTAN

Led the society's operations and managed large-scale national sports events, including the **All Pakistan Sports Festival** hosting universities from across the country. Oversaw **event planning, logistics, and coordination**, including **participant accommodation, catering, scheduling, and venue management**. Served as the **liaison between university administration, sponsors, and teams**, ensuring smooth execution and high participant satisfaction. Strengthened the society's visibility through efficient teamwork, strategic planning, and on-ground leadership.

Worked on the **backend architecture and API development** for computer vision-driven applications. Designed and implemented **RESTful APIs using FastAPI**, integrated with **Swagger UI** for seamless documentation and testing. Contributed to building **scalable, modular backend systems**, focusing on clean architecture, performance, and maintainability. Collaborated with the core engineering team to ensure smooth **data flow between frontend and AI models**, laying the groundwork for efficient deployment and integration of vision-based products.

FREELANCER – 17/04/2021 – Current – TAXILA, PAKISTAN

Delivered end-to-end AI and web solutions for diverse clients, evolving from **building interactive front-end interfaces** to **integrating Agentic AI systems** into production-level applications. Developed and deployed **custom-trained, high-accuracy models** for automation and decision support. Implemented **AI-driven workflows using n8n**, enabling intelligent task orchestration and process automation. Combined expertise in **frontend/backend development and machine learning** to create seamless, scalable, and adaptive digital products that deliver measurable impact.

LANGUAGE SKILLS

Mother tongue(s): **URDU**
Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C1	C1	C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

PROJECTS

Real-Time Traffic Data Engineering Platform

Developed a full-stack data engineering platform to ingest, process and visualise real-time traffic data. Built components including a traffic-data generator, streaming ingestion via Apache Kafka, storage in PostgreSQL and object archive in MinIO, workflow orchestration with Apache Airflow and optional Apache Flink stream processing. Crafted a backend API (Flask) and a React-based dashboard for interactive visualisation of traffic metrics, anomaly detection and trend analytics. Designed the infrastructure using Docker/Docker-Compose for end-to-end deployment, achieving latency of under 5 seconds and throughput of ~100-200 events/minute.

Link <https://github.com/Mortaza76/DE-Project>

Store-Level Sales Forecasting with LSTM

Built a deep-learning solution to forecast item- and store-level sales using an LSTM (Long Short Term Memory) network. Merged historical datasets, engineered time-series and categorical features (store, item, date), normalized data, and designed a two-layer LSTM model with dropout for regularisation. Evaluated performance using RMSE, MAE and R² scores, visualising predicted vs actual sales trends. Delivered outputs as per-item and per-store CSV files and enabled forecasting of the next 7 days of sales for each store-item combination.

Technologies: Python, TensorFlow/Keras, Pandas, Scikit-Learn, Matplotlib.

Link <https://github.com/Mortaza76/LSTM-StorePredictor>

Document Processing Pipeline – Text Extraction

Developed an end-to-end pipeline for extracting structured information from documents (ID cards, insurance cards, medical facesheets). Implemented multi-format ingestion (PDF, JPG, PNG), applied preprocessing (deskewing, denoising, cropping) and used a hybrid extraction strategy combining a multimodal LLM for text + bounding-boxes with traditional OCR engines. Built document classification (type, issuing state), merged results across multiple OCR engines via IoU matching and confidence voting, and extracted normalized fields (e.g., name, DOB, ID number). Exposed the system via a RESTful API for downstream integration.

Technologies: Python, FastAPI, OpenCV, Pillow, pdf2image/PyMuPDF, PaddleOCR, EasyOCR, Tesseract, spaCy (NER), Transformers/ ViT, Dockerized/Modular design.

Link <https://github.com/Mortaza76/Text-Extraction>

Diffusion-Model Art Generator

Developed an optimized version of the Stable Diffusion 2.0 pipeline tailored for Apple Silicon (M1/M2) by leveraging Metal Performance Shaders (MPS) acceleration. Implemented three script variants—basic, prompt-enhanced with fixed seed, and high-quality mode—featuring attention slicing for reduced memory footprint and reproducible results. Enabled users to generate high-fidelity AI-art with fine-tuned prompts and resolutions, achieving efficient inference on Mac hardware. Used technologies such as Python, Diffusers, Transformers, Accelerate and Metal-optimized pipelines.

Link <https://github.com/Mortaza76/diffusion-models>

FaceDetection-DB: Real-Time Face Detection & Database Integration

Designed and implemented a fully operational face-detection system that captures, processes and stores facial data in a structured database. The pipeline includes real-time detection using computer vision techniques, extraction of facial bounding boxes and embeddings, automated storage into a relational/NoSQL face database, and efficient retrieval for applications such as attendance systems or identity verification. Engineered the system for modular integration, including an API layer and database schema optimized for high-throughput and low-latency use-cases.

Technologies: Python, OpenCV/Deep-learning detector (e.g., MTCNN or YOLO-derived), face embedding extraction (e.g., FaceNet/ArcFace), SQL/NoSQL database design, RESTful API, Docker.

Link <https://github.com/Mortaza76/FaceDetection-DB>

MCP for Ollama Local Models – MSSQL

Built a custom Model Context Protocol (MCP) server that enables a local large language model (LLM) to safely query and interact with a Microsoft SQL Server database via natural-language tools. The architecture uses Python (pyodbc, python-dotenv) and Go (mcphost) to expose database operations (e.g., list tables, run queries) as tools for the LLM. Environment configuration (via .env), ODBC driver setup and tool definition are all automated. Demonstrated full pipeline integration between a local LLM (e.g., qwen2.5:7b) and the database infrastructure.

Technologies: Python 3.10+, Go, pyodbc, ODBC Driver 18 for SQL Server, Docker/virtual env, local LLM via Ollama.

Link <https://github.com/Mortaza76/MCPhost-MSSQL-EasyGuide>

Spam Detection for Medical Billing Agent

Developed an AI-driven system to detect fraudulent or spam queries in medical billing workflows. The solution applies document/text extraction and classification to identify invalid submission attempts (e.g., fake practitioner codes, dummy claims) and flags them for review. Built end-to-end: data ingestion, feature engineering (practitioner metadata, query content), model training (high-accuracy classifier), and deployment of the pipeline within a production workflow.

Technologies: Python, NLP classification, document extraction, machine learning model tuning, API integration, backend workflow automation.

Link https://github.com/Mortaza76/Spam-Detection_Medical_Billing_Agent

AI-Powered Automated Hiring Workflow

Built a full-lifecycle recruitment automation platform using n8n and orchestration of multiple AI models to handle applications, scheduling, and communication for a company. Key features include a candidate submission form, acknowledgment email automation, AI-summary of applicant data, decision routing (approve/reject), automated interview slot booking (via Google Calendar + Google Meet), and dynamic personalized email responses for shortlisted or rejected applicants. Integrated multiple language models (e.g., Groq Gemma, Qwen, DeepSeek) and tied the workflow into real-world HR systems. Repository demonstrates a deployable workflow for HR automation.

Technologies: Python, n8n, AI-model integration, REST APIs, Google Calendar API, Gmail/SMTP automation, Docker/DevOps.

Link <https://github.com/Mortaza76/Hiring-Process-Automated>

● SKILLS

Programming & Frameworks

Hugging Face Transformers | OpenCV | LangChain | Python | PyTorch | Flask | LlamaIndex | Model Context Protocol (MCP) | scikit-learn | Fast API

Machine Learning & AI

Supervised & Unsupervised Learning | Deep Learning | Transfer Learning | Fine-tuning LLMs | RAG (Retrieval-Augmented Generation) | Reinforcement Learning | Computer Vision | NLP | Gen AI

Data Science & Analytics

Data Cleaning & Wrangling | Feature Engineering | Exploratory Data Analysis (EDA) | Statistical Modelling | Time Series Forecasting | Predictive Analytics | Data Visualization (Matplotlib, Plotly, Power BI)

MLOps & Deployment

Docker | CI/CD | MLflow | DVC | RESTful APIs | Model Serving | Scalability Optimization | Version Control (Git/GitHub)

Data Engineering & Storage

SQL | NoSQL | Data Lakes | ETL Pipelines | Caching Strategies | Apache Kafka

Automation & Orchestration

n8n | Workflow Automation | API Integration | Agentic AI Systems

Other Tools & Skills

Gradio | Swagger UI | Streamlit | FastAPI | Git | Jupyter | Tensorflow | Prompt Engineering | Context Engineering | AI System Design

Soft Skills

Analytical Thinking | Problem Solving | Leadership | Project Management | Cross-functional Collaboration | Technical Communication